

A Three-Branch Polynomial Hénon Candidate: Exact Primitive-Word and Transfer-Prefix Pilot

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Abstract

We design a three-branch polynomial Hénon candidate for Route A and report an exact, deliberately limited pilot. For $H_a(x, y) = (x^3 - 3x + a - y, x)$ at $a = 1/7$, the cubic has three monotonicity intervals. We freeze their full one-sided symbolic word model and three representative Jacobian samples, enumerate all primitive necklaces through length six, and build a 6×6 matrix-valued transfer prefix. Integer trace decomposition and the finite determinant are independently reproduced and mutation-tested. No complete Hénon coding or Fredholm ownership is assumed: the result is an A1 symbolic pilot and an A2 discrete prefix only.

1 Candidate and evidence boundary

Consider the area-preserving polynomial Hénon map

$$H_{1/7}(x, y) = (x^3 - 3x + 1/7 - y, x), \quad \det DH_{1/7} = 1.$$

The cubic's monotonicity intervals are labeled 0, 1, 2 by $(-\infty, -1)$, $(-1, 1)$, $(1, \infty)$. A Markov partition, invariant compact set, and coding theorem are not supplied here; instead we explicitly freeze the full shift Σ_3 as a falsifiable screening model. At $\xi = (-2, 0, 3)$ the derivative samples are $\sigma = (9, -3, 24)$ and

$$B_j = \begin{pmatrix} \sigma_j & -1 \\ 1 & 0 \end{pmatrix}, \quad \det B_j = 1.$$

These are representative matrices, not asserted periodic-orbit monodromies.

2 Primitive ledger and finite transfer

For a word $w \in \{0, 1, 2\}^n$, let $[w]$ be its lexicographically least cyclic rotation, retaining it only when w is not a repetition. Put $B_w = B_{w_{n-1}} \cdots B_{w_0}$. The block transfer matrix A has 2×2 blocks $A_{ij} = B_j$ for every $i, j \in \{0, 1, 2\}$. Direct integer enumeration gives the following primitive-necklace counts.

length n	1	2	3	4	5	6	total
$ \mathcal{P}_n $	3	3	8	18	48	116	196

Proposition 1 (finite trace decomposition). *For $1 \leq n \leq 6$,*

$$\mathrm{Tr}(A^n) = \sum_{d|n} d \sum_{[w] \in \mathcal{P}_d} \mathrm{Tr}(B_w^{n/d}).$$

Proof. Expand the block trace as a sum over closed length- n symbol paths. Each path is a repetition of a unique primitive necklace of length $d \mid n$; there are d distinguished starts, and cyclic invariance of trace identifies the corresponding matrix products. The certificate checks the equality for all six values of n with integer arithmetic. $\square$

For this frozen sample, the direct trace vector is

$$(\mathrm{Tr} A, \dots, \mathrm{Tr} A^6) = (30, 882, 26190, 777762, 23097150, 685914642), \quad D(z) = 1 - 30z + 9z^2.$$

Writing $D(z) = \det(I - zA) = \sum c_k z^k$, SymPy independently verifies

$$kc_k = - \sum_{j=1}^k c_{k-j} \mathrm{Tr}(A^j), \quad 1 \leq k \leq 6,$$

with zero coefficients after the actual degree. The observed degree-two collapse is a diagnostic of the rank-one block construction, not a spectral claim.

3 Route-A assessment and reproducibility

The canonical evidence file records all 196 rows, branch counts, matrices, repeated traces, six direct traces, primitive decompositions, and determinant coefficients. An independent checker, a SymPy/Newton cross-check, canonical replay, and nine hostile mutations all pass. The finite matrix is not called a Fredholm operator: no function space, nuclearity estimate, geometric orbit completeness, or zero-count experiment is claimed. Accordingly, the four-layer status is recorded as

$$\begin{aligned} A1 &= A1_OPEN \text{ (symbolic-pilot qualification),} \\ A2 &= A2_CERTIFIED_PREFIX \text{ (discrete-prefix qualification),} \\ A3 &= A3_NOT_ADDRESSED, \quad A4 = A4_FAIL. \end{aligned}$$

No arithmetic/local data, Euler factors, root numbers, automorphy, Hilbert–Pólya operator, or Route-B authorization is present. The next experiment must freeze a compact candidate set and independently solve the actual Hénon periodic equations before upgrading A1.